Phase 1: Problem Identification & Proposal

Problem Statement

Digital images often suffer from blur due to various factors such as:

- Camera shake
- Motion of objects
- Low-light conditions
- Defocus during capture

Blurred images reduce visual quality and can negatively affect:

- Medical diagnosis
- Surveillance systems
- Autonomous driving
- Face recognition systems
- Document digitization

Objective:

To develop a deep learning-based model using Generative Adversarial Networks (GANs) that can restore sharp images from blurred inputs by learning the mapping between blurred and sharp image pairs.

Motivation and Relevance

Why This Problem Matters

1. Real-world applications

- CCTV footage enhance ement

- Satellite imagery improvement
- Smartphone camera post-processing
- Medical imaging clarity
- OCR systems for scanned documents

2. Limitations of traditional methods

- Classical deconvolution methods require blur kernel estimation.
- Sensitive to noise.
- Not robust for complex real-world blur.

3. Why Deep Learning?

- Learns blur patterns directly from data.
- Handles non-uniform and complex motion blur.
- Produces perceptually realistic outputs.

4. Why GANs Specifically?

- GANs generate sharper and more visually appealing images.
- Traditional CNN models minimize pixel-wise loss → produce smooth/blurry outputs.
- GANs optimize perceptual realism using adversarial loss.

Type of Data to be Collected

Dataset Requirements

Blurred Image (Input)

Corresponding Sharp Image (Ground Truth)

Expected Deep Learning Approach

Model Architecture: GAN-Based Deblurring

Using a Conditional GAN (cGAN) architecture.

Architecture Overview

The system consists of two networks:

1. <u>Generator (G)</u>
   - Takes blurred image as input
   - Outputs deblurred (restored) image
   - Likely architecture:
     - U-Net based encoder-decoder
     - Residual blocks
     - Skip connections
2. <u>Discriminator (D)</u>
   - Distinguishes between:
     - Real sharp image
     - Generated deblurred image
   - PatchGAN discriminator can be used

Training Objective

The model will optimize a combination of:

1. <u>Adversarial Loss</u>

   - Encourages realistic image generation.

2. <u>Content Loss (L1 or L2 Loss)</u>
   - Ensures pixel-level similarity.

Loss Function (Combined)

$$Total\ Loss = Adversarial\ Loss + \lambda 1 L1\ Loss + \lambda 2 Perceptual\ Loss$$

Implementation Framework

- Python
- TensorFlow / PyTorch
- GPU acceleration

Evaluation Metrics

To measure performance:

- PSNR (Peak Signal-to-Noise Ratio)
- SSIM (Structural Similarity Index)
- Visual comparison

Expected Outcome

A trained GAN model capable of:

- Restoring blurred images
- Producing perceptually sharp outputs

Quantitative improvement in PSNR & SSIM

Demonstration of GAN superiority over basic CNN